IFT6135 - Assignment 3 Report

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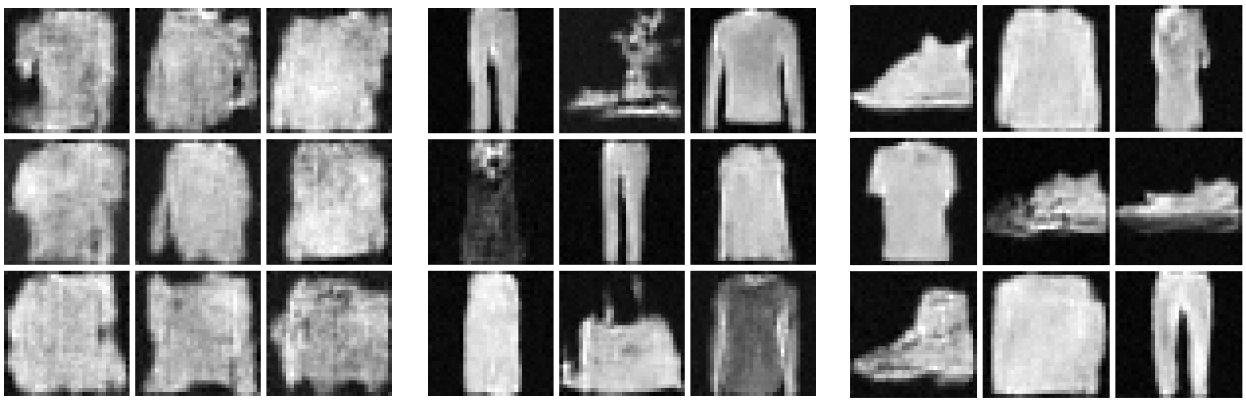
Problem 1: Denoising Diffusion Probabilistic Models

Question 8: Training progression and sample quality



(a) Epoch 2 (b) Epoch 10 (c) Epoch 18

Figure 1: DDPM unconditional samples (3×3 grids at 32×32) drawn from the model at three points during training.

Sample sharpness

At epoch-2, the model has only seen the dataset twice and the reverse chain returns rough rectangular blobs of uniform luminance, with no recognizable garment structure. After only two epochs, the noise predictor seems to have learned the marginal pixel statistics and the bulk shape prior (objects on a dark background, roughly square footprint) but not yet category-specific structure.

By epoch-10, most cells in the grid contain unambiguous FashionMNIST silhouettes (trousers, long-sleeve top, dress, skirt, knit pullover), with crisp object and background separation, and only mild blur at the edges. By epoch-18, edges seem sharper, but many samples are still a bit weird looking with difficulties with sleeves. The differences between epoch-10 and 18 don't seem to be too big.

Diversity

The epoch-18 grid alone covers at least seven of the ten FashionMNIST categories (sneaker, ankle boot, pullover, t-shirt, trousers, sandal/slipper, heeled shoe, dress) across nine cells, which suggests the model has not collapsed to a small subset of modes.

Common failure modes

- **Fractal textures:** A small fraction of samples (e.g. the top-middle cell at epoch 10) seems to contain noise-like or fractal-looking textures rather than a recognizable garment. These look like trajectories that the reverse chain failed to commit to any nearby data mode.

- **Residual blur:** Even at epoch-18, large flat regions (the body of a sweater, the leg of a trouser) remain perceptibly softer than real 32×32 FashionMNIST images. This is the expected price of the simplified L_2 objective, which regresses against the mean of plausible completions.
- **Asymmetric or truncated extremities:** Several cells have one sleeve cleanly resolved and the other faded, broken, or noticeably thinner (e.g. the left sleeve of the dress at the top-right of the epoch-18 grid). Where the dataset has multiple plausible sleeve completions for a given silhouette, the L_2 noise predictor seems to average over them and commit to a compromise that visibly mismatches the surrounding garment.

Practical improvements

- **Cosine β schedule:** The linear schedule used here drops the signal-to-noise ratio to near zero across most of the chain, so the network spends most of its training budget on timesteps where there is essentially no signal left to predict. The cosine schedule keeps the signal-to-noise ratio bounded away from zero through more of the chain, which could help reduce the residual blur observed above by giving the network more useful gradient signal at intermediate t.
- **Longer training:** The training loss was still trending down at the end of 20 epochs, so simply training for longer should provide additional quality, with no architectural change required.
- **Deeper or wider U-Net backbone:** Adding base channels, residual blocks per resolution, or self-attention layers at the bottleneck would increase the network's capacity to synthesize high-frequency texture in the structure-forming regime near $t = 0$, which is where most of the perceptual quality is decided.
- **Learned reverse variance:** The current implementation fixes $\sigma_t^2 = \beta_t$ at every reverse step. Letting the network also predict $\sigma_\theta(\mathbf{x}_t, t)$ could improve both negative log-likelihood and sample quality, since the optimal reverse variance lies between two natural bounds rather than at either endpoint.

Question 9: Intermediate denoising stages

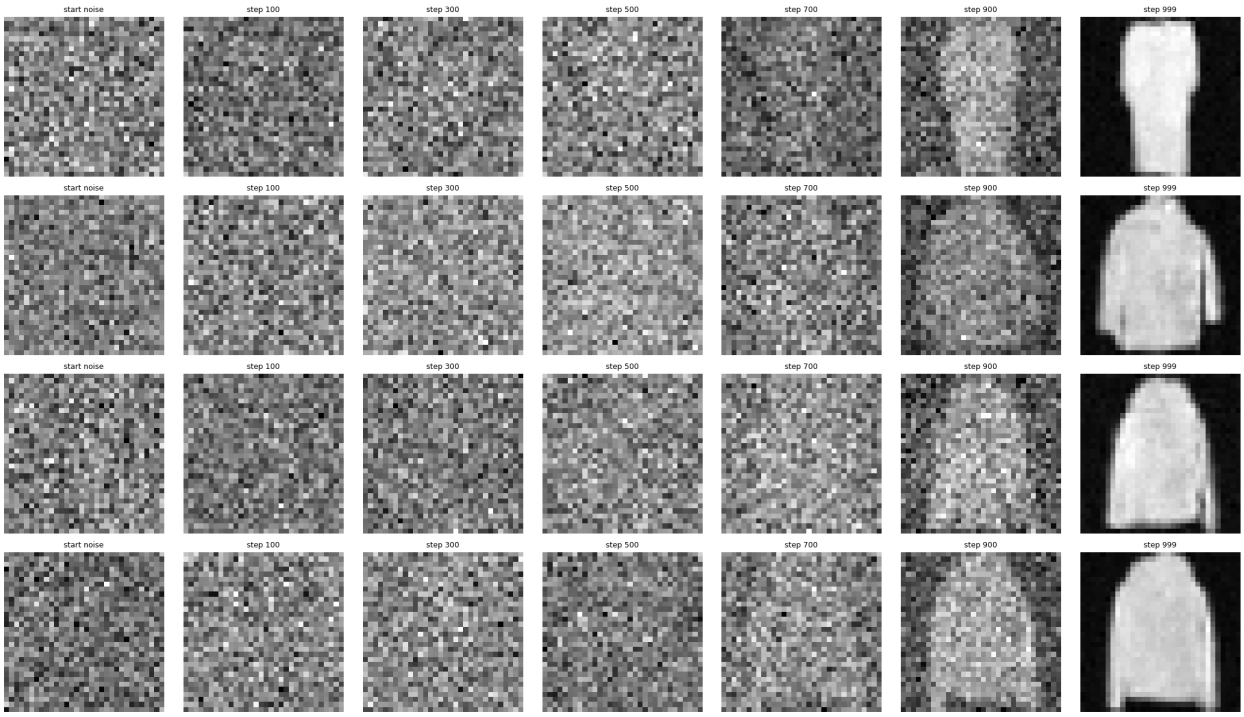


Figure 2: Four reverse trajectories, captured at the same six points along the 1000-step reverse chain. All four runs use the same trained network.

At "start noise" the iterate is pure Gaussian noise, consistent with the prior $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ that opens the generative process, and with the iterative-denoising trajectory $\mathbf{x}_T \to \cdots \to \mathbf{x}_0$. Through the first $\sim$700 reverse steps, the panels at step 100, 300, 500, and 700 are visually almost indistinguishable from the starting noise. The signal weight $\sqrt{\bar{\alpha}_t}$ in the forward marginal is essentially zero under the linear β schedule at high t, so the signal contribution to $\mathbf{x}_t$ is absent and the noise prediction $\epsilon_\theta(\mathbf{x}_t, t)$ is on average close to the input itself. Inside the reverse-mean update μ_θ from the problem statement, the noise-prediction correction is scaled by the factor $\beta_t/\sqrt{1-\bar{\alpha}_t}$, which is small early in the chain ($\sqrt{1-\bar{\alpha}_t} \approx 1$ and β_t is bounded by the schedule), so the correction barely moves the iterate and the update mostly subtracts variance without incorporating structure. Visually, this regime is just noise that becomes slightly less energetic per step.

A silhouette becomes recognizable only around step 900. The transition is rapid because $\sqrt{\bar{\alpha}_t}$ grows quickly once the steps are past about 700, and the factor $\beta_t/\sqrt{1-\bar{\alpha}_t}$ grows alongside it as $\sqrt{1-\bar{\alpha}_t}$ shrinks, so the noise-prediction correction starts to dominate the update and pulls the iterate toward a low-frequency object outline (a body, sleeves, the rough shape of a dress or trouser leg). At this stage the high-frequency texture is still missing, what is visible is the silhouette and broad shading. This concentrates the phase where structure forms in a small window near $t = 0$, which seems to be the practical reason DDPM needs roughly $T = 1000$ steps to sample well.

The simplified L_2 training loss treats every t identically. The progression to fine details is therefore not an explicit objective, it is forced by the geometry of the schedule with $\bar{\alpha}_t$. Near the end of the chain the reverse-step variance σ_t^2 shrinks to near zero, so $\mathbf{x}_{t-1} \approx \mu_\theta(\mathbf{x}_t, t)$ and the reverse step becomes a near-deterministic refinement that adds the high-frequency detail buried below the noise floor at earlier t. Most of the chain therefore spends its budget on coarse noise removal and only the final tail on high-frequency synthesis.

Problem 2: Flow Matching

Question 6: Theoretical comparison with DDPM

Training objectives

Both objectives are simple L_2 regressions, but they regress against different quantities on different time domains. DDPM regresses against the injected noise ϵ on a discrete time grid $t \in \{0, \ldots, T-1\}$ with per-step weighting implicitly defined by the $\bar{\alpha}_t$ schedule. Flow Matching regresses against the constant-velocity target on the continuous interval $t \in [0, 1]$, with no schedule-dependent weighting. The DDPM target therefore depends on the noise schedule via $\bar{\alpha}_t$, while the FM target is a constant displacement along a straight path and decoupled from any noise schedule.

Stochastic vs. deterministic generation

DDPM sampling is stochastic. Each reverse transition is a Gaussian with a fresh noise injected at every step, so two runs from the same starting noise $\mathbf{x}_T$ already produce different samples. Flow Matching sampling integrates the deterministic FM ODE driven by the learned velocity field with no noise injection, so the same starting noise always maps to the same sample. The practical consequences are that FM trajectories are reproducible and usable by high-order solvers like Midpoint, while DDPM needs more reverse steps to average out the stochastic component.

U-Net usage

The U-Net architecture is identical in both problems. The differences are in what the network predicts, how time is presented to it, and how its output is composed at sampling time. DDPM uses it as $\epsilon_\theta(\mathbf{x}_t, t)$ conditioned on a discrete timestep index, with the output interpreted as the noise component to subtract inside the reverse mean from the problem statement. Flow Matching reuses the same network as $\mathbf{v}_\theta(\mathbf{x}_t, t)$ on a continuous time $t \in [0, 1]$ that is rescaled to the same integer embedding range via $\lfloor t \cdot 1000 \rfloor$, with the output used directly as a velocity inside the Euler / Midpoint update.

Question 7: Sampling quality and efficiency

Results

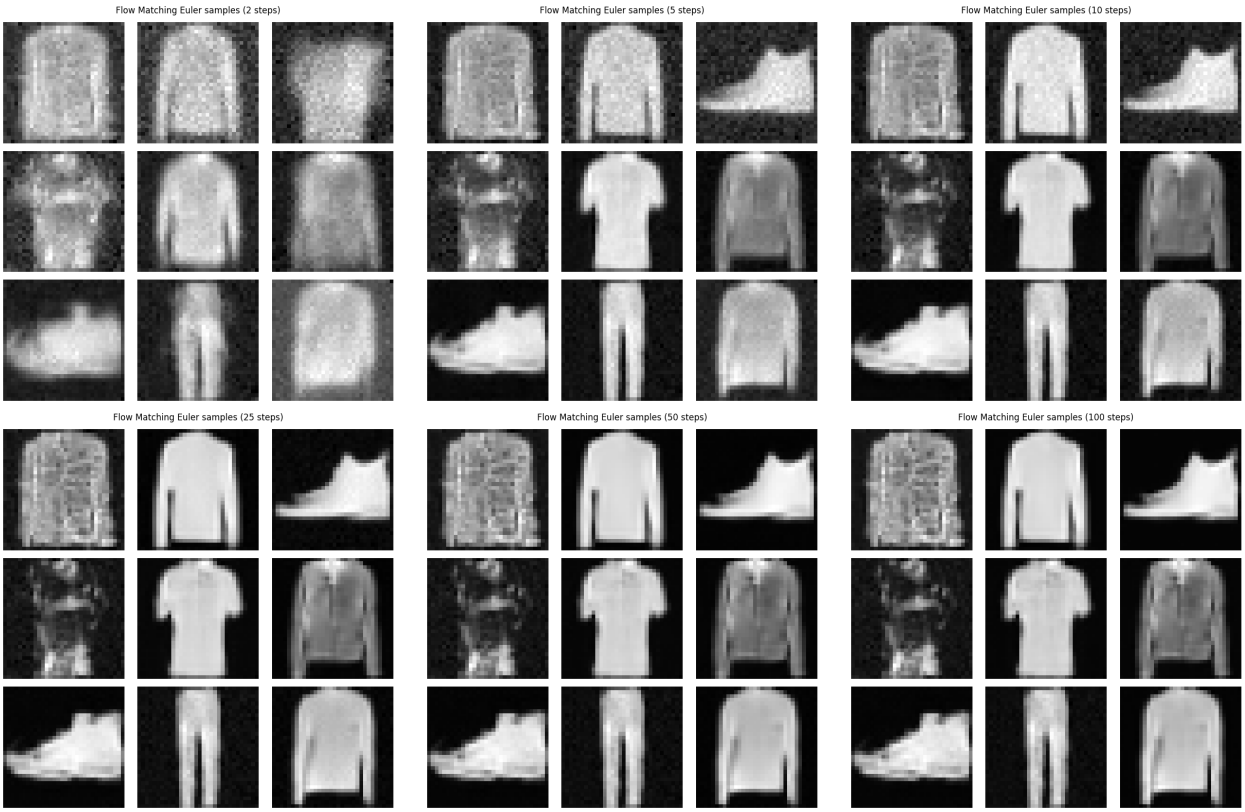


Figure 3: Flow Matching samples under Euler integration at $\{2, 5, 10, 25, 50, 100\}$ steps. Each panel reuses the same fixed starting-noise tensor, and only the number of Euler updates changes.

Effect of step count on Euler

At 2 steps Euler produces only blurry blobs with rough garment outlines but no reliable category identity. Past 25 steps the marginal improvement from adding more Euler steps is imperceptible. The 25, 50, and 100-step rows look nearly indistinguishable. This appears to match that for a well-trained, sufficiently smooth velocity field the Euler discretization converges fast.

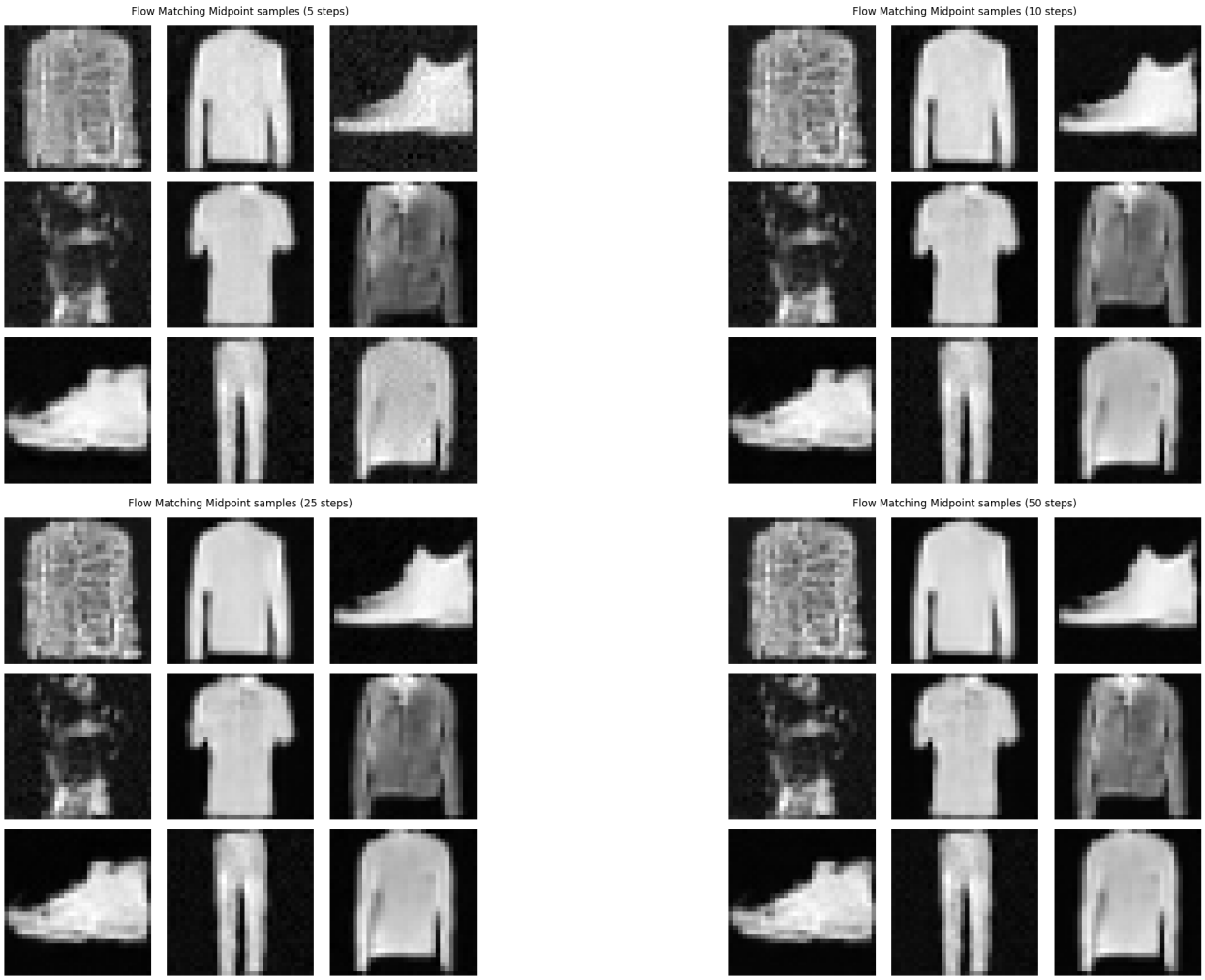


Figure 4: Flow Matching samples under Midpoint integration at $\{5, 10, 25, 50\}$ steps. Same fixed starting noise as Figure 3. Midpoint uses two velocity evaluations per step, so the NFE budgets are $\{10, 20, 50, 100\}$.

Effect of step count on Midpoint

Midpoint reaches usable quality much earlier. At 5 steps (NFE = 10) the samples are already competitive with Euler at 25 steps (NFE = 25), and at 10 steps (NFE = 20) they essentially match Euler at 100 steps. Beyond 10 steps the Midpoint samples are also visually saturated. At matched NFE budgets, Midpoint's second-order accuracy seems to be the deciding factor at small budgets and becomes irrelevant at large budgets, which is the regime where a higher-order ODE solver is expected to help.

Comparison with DDPM at matched starting noise

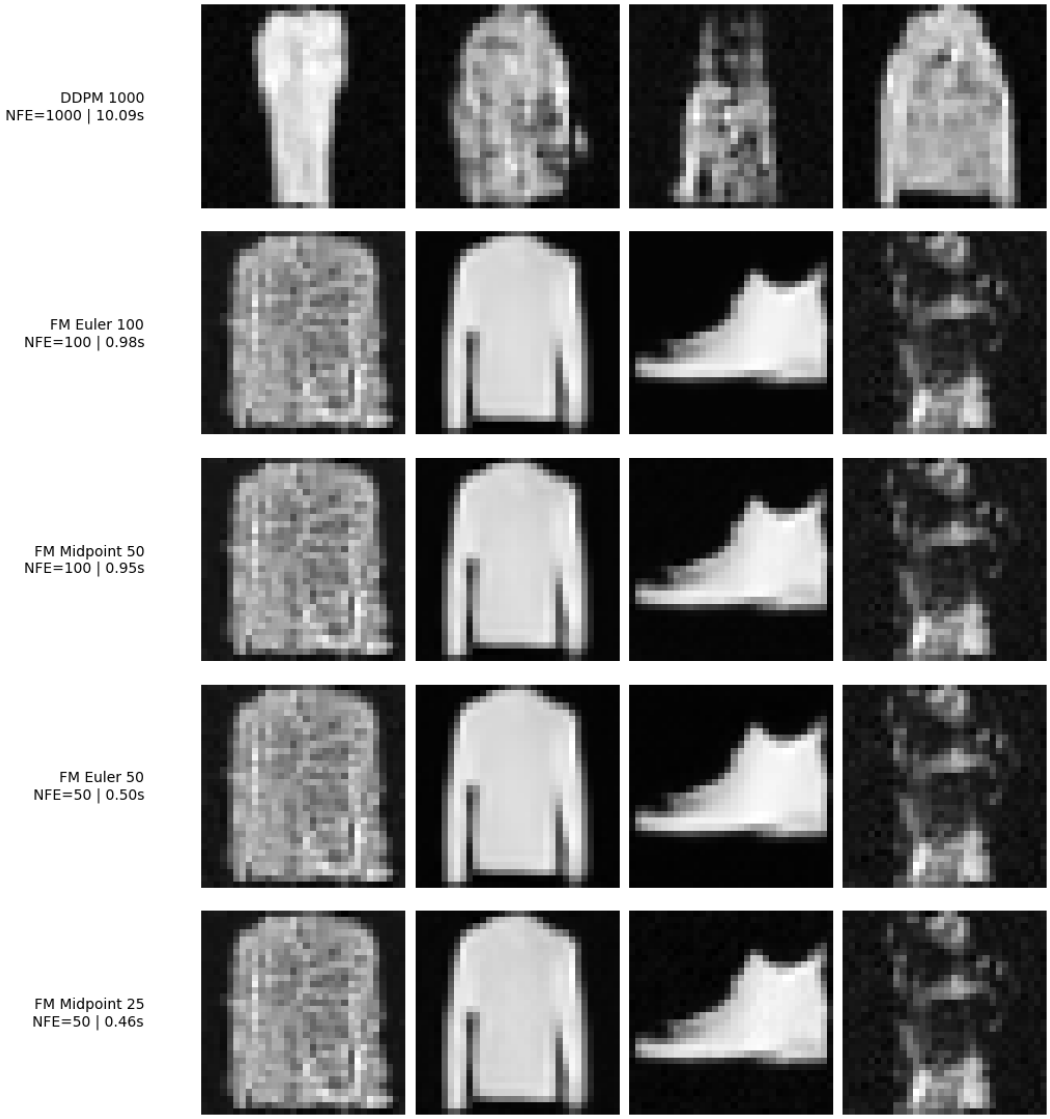


Figure 5: DDPM (1000-step stochastic reverse chain) compared with Flow Matching at four (sampler, step count) configurations. All five configurations use the same starting noise tensor. The per-row label reports the sampler, step count, NFE, and wall-clock runtime on an RTX 3060.

Table 1: Network-evaluation budgets and runtimes for Figure 5. Wall-clock scales linearly with NFE within FM, and runtime is dominated by the U-Net forward pass, so the Midpoint and Euler runs at matched NFE finish in essentially identical time.

Configuration	Steps	NFE	Runtime (s)
DDPM (1000 steps)	1000	1000	10.09
FM Euler 100	100	100	0.98
FM Midpoint 50	50	100	0.95
FM Euler 50	50	50	0.50
FM Midpoint 25	25	50	0.46

Even with the same starting noise tensor, DDPM and FM produce visibly different samples (Figure 5, column-by-column). This is the expected consequence of the stochastic-versus-deterministic distinction discussed in Question 6, since DDPM and FM use the same starting noise for entirely different roles (the prior $\mathbf{x}_T$ of a noisy Markov chain on one side, and the boundary condition $\mathbf{x}_1$ of a deterministic ODE on the other).

Within the FM rows the four sampler configurations produce nearly identical samples, which suggests both that the trained velocity field is converged and that NFE = 50 with Midpoint is already on the plateau.

Runtime / NFE tradeoff

Table 1 makes the practical tradeoff explicit. At equal sample quality (DDPM vs. FM Midpoint 50), Flow Matching is roughly 10.6× faster in wall-clock and uses 10× fewer network evaluations. At matched NFE budgets (Euler 100 vs. Midpoint 50, or Euler 50 vs. Midpoint 25), the two FM samplers are within a few hundredths of a second of each other. Runtime is essentially a function of NFE alone, but Midpoint's second-order accuracy lets it use half as many integration steps. The cheapest configuration that still produces clean samples here is FM Midpoint 25 (NFE = 50) at 0.46 s, roughly a 22× speedup over the 1000-step DDPM baseline at indistinguishable visual quality.

Trajectory comparison with DDPM

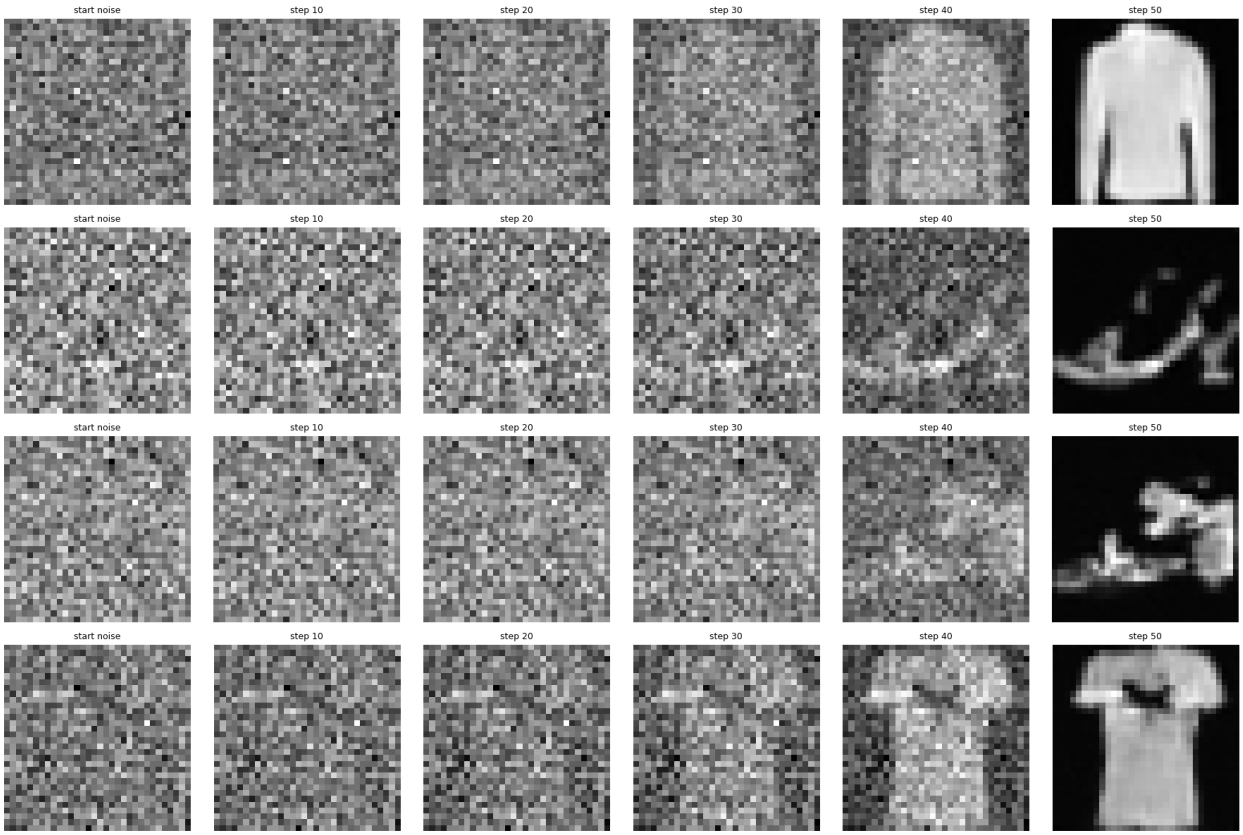
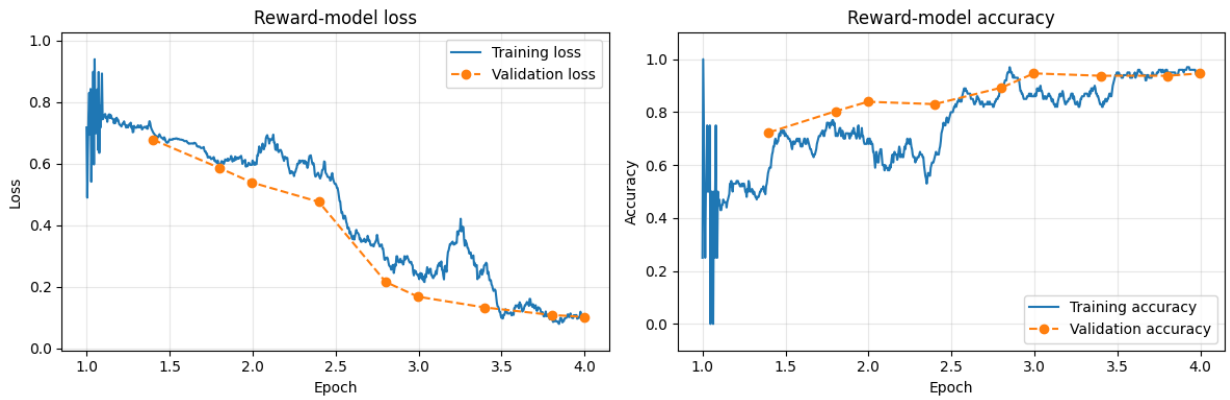


Figure 6: Flow Matching ODE trajectory under Midpoint with 50 steps. Column labels record the number of completed integration steps from $t = 1$ towards $t = 0$.

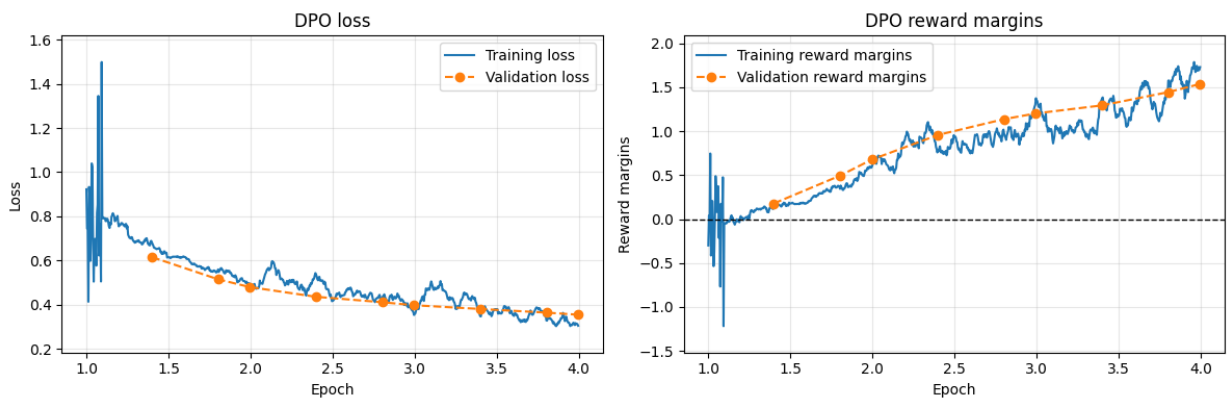
The FM trajectory in Figure 6 mirrors the qualitative behaviour seen for DDPM in Figure 2. The iterate looks like noise for most of the chain and snaps to a recognizable garment only over the last 10–20% of the integration. The mechanism is different, but the practical implication seems similar, with the structure-forming phase concentrated near the data endpoint of the path.

Problem 3: Language Model Alignment

Training summary



(a) Reward-model training, showing Bradley–Terry loss and pairwise accuracy.



(b) DPO training, showing loss and implicit reward margin on chosen minus rejected.

Figure 7: Training and validation curves for Problem 3.

Both runs converge cleanly. The reward model reaches a held-out preference accuracy of 94.6% (loss 0.10), with the validation loss dropping from 0.69 to 0.10 over the 3 epochs. The DPO policy reaches a held-out preference accuracy of 91.1% with a mean implicit reward margin of 1.54. The reward margin grows roughly monotonically from 0 to ≈ 1.54 during DPO training, which suggests that the policy is consistently assigning higher likelihood than the reference to chosen responses.

Question 9: DPO vs PPO theoretical comparison

Optimizing directly from preference pairs

PPO-based RLHF first fits an external reward model r_ϕ from the preference data, then optimizes the policy with on-policy reinforcement learning against r_ϕ subject to a KL penalty to π_{ref}. The explicit objective is

$$\max_{\pi} \mathbb{E}_{\mathbf{y}\sim\pi}[r(\mathbf{x},\mathbf{y})] - \beta\,\mathrm{KL}\big(\pi(\cdot \mid \mathbf{x}) \,\|\, \pi_{\text{ref}}(\cdot \mid \mathbf{x})\big).$$

DPO derives a single supervised loss on offline preference pairs that is equivalent to optimizing this objective at the optimum. First, setting the Lagrangian's functional derivative to zero and enforcing the normalization

constraint $\sum_{\mathbf{y}} \pi(\mathbf{y} \mid \mathbf{x}) = 1$ gives the policy

$$\pi^\star(\mathbf{y} \mid \mathbf{x}) = \frac{1}{Z(\mathbf{x})}\, \pi_{\text{ref}}(\mathbf{y} \mid \mathbf{x})\, \exp\!\Big(\tfrac{1}{\beta}\, r(\mathbf{x}, \mathbf{y})\Big)\,,$$

with prompt-conditioned function $Z(\mathbf{x}) = \sum_{\mathbf{y}} \pi_{\text{ref}}(\mathbf{y} \mid \mathbf{x})\, \exp(r(\mathbf{x}, \mathbf{y})/\beta)$.

Taking the log of the closed form and rearranging to express the reward in terms of the policy,

$$r(\mathbf{x}, \mathbf{y}) = \beta\, \log \frac{\pi^\star(\mathbf{y} \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y} \mid \mathbf{x})} + \beta\, \log Z(\mathbf{x}).$$

Plugging this expression for r into Bradley-Terry expression yields

$$P(\mathbf{y}_w \succ \mathbf{y}_l \mid \mathbf{x}) = \sigma\!\left(\beta\, \log \frac{\pi^\star(\mathbf{y}_w \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_w \mid \mathbf{x})} - \beta\, \log \frac{\pi^\star(\mathbf{y}_l \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_l \mid \mathbf{x})}\right),$$

since the $\beta \log Z(\mathbf{x})$ terms cancel in the difference.

Treating the unknown optimum $\pi^\star$ as a trainable policy π_θ, the negative log-likelihood of the observed preference pairs is exactly the DPO loss from the problem statement,

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l)}\left[\log \sigma\!\left(\beta\!\left(\log \frac{\pi_\theta(\mathbf{y}_w \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_w \mid \mathbf{x})} - \log \frac{\pi_\theta(\mathbf{y}_l \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_l \mid \mathbf{x})}\right)\right)\right]. \tag{1}$$

The result is a binary cross-entropy on the reward between the chosen and rejected responses, so it can be optimized with ordinary backpropagation on offline pairs. The gradient at each pair pushes $\log \pi_\theta(\mathbf{y}_w)$ up and $\log \pi_\theta(\mathbf{y}_l)$ down, so updates are large for pairs the policy currently ranks incorrectly and small for pairs already correctly ranked. There is no rollout policy, no on-policy sampling, and no separately trained r_ϕ, and so it optimizes effectively from preference pairs only.

Role of the frozen reference model and the limit $\beta \to 0$

The reference model π_{ref} acts as an anchor for the policy. In Equation 1, π_{ref} only appears inside the log-ratios $\log[\pi_\theta(\mathbf{y})/\pi_{\text{ref}}(\mathbf{y})]$ that sit inside σ, so the DPO loss only registers deviations of the policy from the reference rather than absolute likelihoods. The coefficient β multiplies the entire log-ratio difference inside σ and sets how strict that anchor is. Larger β keeps the policy close to the reference and slows learning. Smaller β relaxes the anchor and lets the policy move further per step.

If $\beta \to 0$ (or the reference is removed entirely), the anchor disappears. Equation 1 is then minimized only by sending the log-ratio margin between chosen and rejected responses to $+\infty$, with nothing pulling the policy back toward π_{ref}. The policy can pile arbitrarily large probability on the chosen responses and arbitrarily little on the rejected ones. In practice this means the policy loses the natural writing style it picked up during SFT, any mistakes the human labelers made get amplified instead of smoothed out, and the policy starts picking up on accidental patterns in the training pairs rather than on what the labelers actually cared about (reward hacking). PPO controls it with an explicit KL penalty, and DPO bakes the same effect into its loss through the log-ratio.

Advantage and disadvantage relative to PPO

Advantage: DPO is simpler to set up and run than PPO. It trains one model with a regular supervised loss on a fixed dataset of preference pairs. PPO instead needs a separately trained reward model, a loop that generates and scores fresh responses during training, and a value network to estimate baselines for the RL update. With fewer moving parts, DPO has just one main hyperparameter (β) to tune instead of the many PPO needs, which makes it more stable and easier to reproduce across runs.

Disadvantage: DPO can only learn from the preference pairs it sees during training. Once those are fixed, it has no way of asking the reward signal about new responses, so prompts or response styles outside the training data get no useful update. PPO does not have this problem because it keeps generating fresh

candidates from the current policy at each step and scoring them with the reward model, which lets it correct itself on responses the static dataset never contained. Iterative DPO, we have seen, closes most of the gap by periodically re-sampling preference pairs from the current policy, recovering the on-policy property of PPO without the RL part.

Question 10: Best-of-N analysis

Mean reward as a function of N

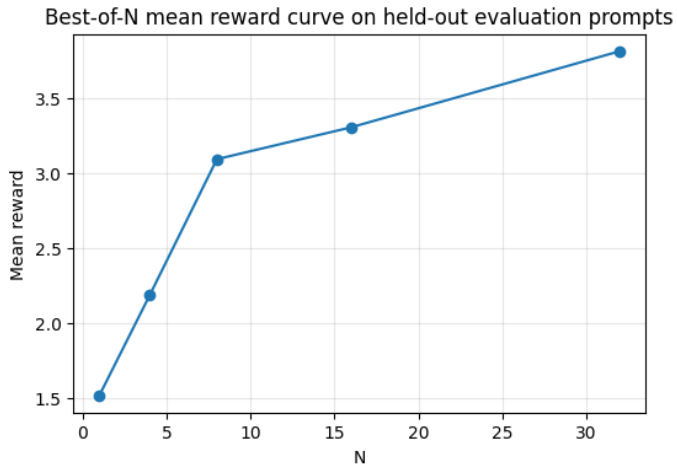


Figure 8: Best-of-N mean reward over 5 held-out evaluation prompts, $N \in \{1, 4, 8, 16, 32\}$. The curve is monotone and concave in N with diminishing returns past $N \approx 16$.

From figure 8, the mean reward of the Best-of-N selection on the held-out prompts grows monotonically with N, from ≈ 1.5 at $N = 1$ to ≈ 3.8 at $N = 32$. The steepest gain is between $N = 1$ and $N = 8$ ($\approx +1.6$), and there are clearly diminishing returns past $N = 16$.

The shape of the curve is what one would expect from picking the maximum out of a larger and larger pool of candidates. Doubling N doubles the chances of hitting an unusually good candidate, but the gain shrinks fast because most of the easy improvement comes from avoiding the worst few candidates and the rest is rarer luck. Doubling N from 16 to 32 only buys about $+0.5$ reward at twice the inference cost.

Best-of-1 vs. Best-of-16 on a held-out prompt

For the prompt *"Explain what a sonnet is."*, the same DPO policy produces these two candidates.

- **Best-of-1** (reward 3.78): *"A sonnet is a type of poetic composition in which the primary subject is a child or child-like figure. It typically begins with the words 'I am' or 'I sing.'"*
- **Best-of-16** (reward 6.37): *"A sonnet is a piece of music composed for the sole purpose of expressing an idea. It typically begins with a strong, melodic voice, which typically changes to a more verse or a more lyrical voice, typically in the second or"*

Does higher reward match human judgment?

Not on this prompt. A sonnet is a 14-line poem with a fixed rhyme scheme. Both candidates are factually wrong, and the Best-of-16 answer is arguably worse. It asserts that a sonnet is "a piece of music" rather than a poem, and trails off mid-sentence at the 128-token cap. Yet it scores nearly twice as high as Best-of-1.

The reward model is just an approximation of what a human would prefer. As N grows, Best-of-N searches harder and harder for the candidate that the reward model rates highest, which is not the same thing as the candidate a human would actually prefer. This is a reward hacking failure, where optimizing a

proxy hard enough makes its score go up while the true quality stagnates or even goes down. A few specific reward-model limitations show up here:

- The reward model seems to prefer longer responses with a confident, instruction-tuned tone even when they are factually wrong.
- The reward model is a small GPT-2 fine-tuned on about a thousand preference pairs and is never told what is actually true about the world, so it cannot really tell a correct definition apart from a confident-sounding wrong one.
- Literary definitions are probably underrepresented in this small training set, so on a prompt about sonnets the reward model has nothing solid to fall back on and just picks whichever surface form looks the most polished.

The practical takeaway is that Best-of-N amplifies whatever biases the reward model already has. The quality ceiling is set by the reward model itself, and on prompts where the reward model is unreliable, larger N can actively make things worse.